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The future of Escherichia coli vaccines in prevention of recurrent Urinary Tract Disease based on the newest guidelines worldwide.

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Abstract

Introduction:

Urinary tract infections (UTIs) represent a major global health burden, with recurrent UTIs (rUTIs) severely impacting patient quality of life and imposing billions of dollars in annual economic costs. Rising antimicrobial resistance among uropathogenic *Escherichia coli* (UPEC)—the primary causative pathogen—has created an urgent need for non-antibiotic prophylactic strategies, accelerating interest in immunotherapeutic interventions and vaccines.

Aim: The aim of this narrative review is to evaluate E.coli immunoprophylaxis in rURI treatment according to the newest worldwide recommendations and synthesise further

Material and methods:

A narrative literature search was conducted using PubMed and Google Scholar for English- and Polish-language studies published up to May 2026. Keywords included variations of "OM-89", "Uro-Vaxom", "E. coli immunotherapy", and "recurrent UTI". Eligible literature comprised randomized controlled trials, observational studies, systematic reviews, meta-analyses, preclinical models, and official clinical guidelines. A narrative synthesis was performed across core thematic domains.

Results:

UPEC relies heavily on type 1 fimbrial adhesin (FimH) to adhere to the bladder uroepithelium and form intracellular bacterial communities (IBCs) that evade antibiotic clearance and host defenses. Clinically, the oral immunomodulating bacterial extract Uro-Vaxom (OM-89) has historically demonstrated reductions in rUTI frequency and dysuria symptoms. However, recent evaluations highlight study heterogeneity and inconsistent outcome measures, leading international bodies such as the European Association of Urology (EAU) to downgrade their recommendation for OM-89 from strong to weak in 2026. Simultaneously, novel subunit vaccines targeting the FimH adhesin (e.g., adjuvanted FimCH) are emerging; early clinical trials demonstrate a favorable safety and immunogenicity profile, establishing a pathway for upcoming Phase 2 evaluation.

Conclusions:

While OM-89 has long served as a key non-antibiotic prophylactic option, shifting guideline recommendations emphasize the necessity for standardized, high-quality randomized controlled trials to definitively establish its efficacy. Concurrently, target-specific interventions like FimH subunit vaccines represent a promising next-generation immunotherapeutic approach to break the cycle of rUTIs without driving antimicrobial resistance.

Keywords: Recurrent Urinary Tract Infection (rUTI), *Escherichia coli*, Uro-Vaxom (OM-89), FimH adhesin, Vaccine, Immunotherapy, Antimicrobial Resistance.

Introduction

Urinary tract infections (UTIs) are among the most common bacterial diseases worldwide, being around 40% of all hospital-acquired infections and 10-20% non-hospital acquired infections [1]. The European Association of Urology (EAU) classifies UTIs as either complicated (males or women with risk factors or anatomical urinary tract variations) or uncomplicated [2]. The estimated prevalence of uncomplicated UTIs exceeds 50% in the women population, while 20% to 40% of those cases are experiencing recurrent UTIs (rUTIs), defined as at-least two episodes of cystitis in a period of 6 months, or at-least three episodes over 12 months [3,4].

While an isolated episode of acute, simple cystitis, the most common UTI, is considered rather innocuous, recurrent UTIs significantly reduce patients' quality of life [5]. The disease creates an enormous financial and economic burden on the entire society, as it greatly interferes with normal daily activities and reduces patients' work productivity. In the USA alone, the societal costs of management of UTIs is estimated from 2.4 billion up to even 5 billion US dollars annually [3]. According to a study conducted by [Yining He](#) and co. in 2021, the global trend of increasing UTIs cases is projected to continue until at least 2050 [6].

Empirical antibiotics remain the first-line treatment for cystitis, with uropathogenic *Escherichia coli* (UPEC) identified as the leading pathogene, responsible for around 80% of uncomplicated and 65% of complicated UTIs [5, 7]. While most UTIs respond effectively to antibiotics, multidrug-resistant bacteria are being more and more frequently found in urine cultures. Growing antibiotic resistance is an increasing problem addressed by the World Health Organization since 2015. A global action plan to create new and alternative strategies to fight bacterial infections created that year was evaluated and improved over the years, the newest addition in May of 2026. The improved strategy for 2026-2036 emphasizes the need to intensify infection prevention and the overall need for antimicrobials [8]. Following those recommendations, modern urological research has turned toward immunotherapeutic interventions focusing on the primary pathogen virulence.

Uropathogenic *Escherichia coli* utilizes virulent structures; most notably the FimH adhesin found on type 1 pili; to anchor itself tightly to the bladder mucosa, avoiding mechanical clearance of urine flow and forming resilient reservoirs that result in frequent recurrences even

after proper antibiotic treatment [9, 10]. The need to break this cycle of recurrence without further fueling drug resistance resulted in creation of *Escherichia coli* vaccine. There is currently only one UPEC vaccine - Uro-vaxom - available on the market, while few others, targeting especially immunogenic properties of FimH, are currently under study. This review aims to synthesize the current literature on newest guidelines of *E. coli* vaccines mileage in rUTIs treatment.

MATERIALS AND METHODS

This narrative review summarizes current evidence on the role of *E. coli* vaccines and the newest guidelines of its usage in recurrent Urinary Tract Infections treatment. A literature search was conducted in PubMed and Google Scholar for studies published up to May 2026 using keywords related to: “OM-89”, “Uro-vaxom”, “*Escherichia coli* immunoprophylaxis” or “*E.col* immunotherapy”, “Urinary Tract infections” or “recurrent Urinary Tract infections” or “recurrent UTI”.

Only English-language and Polish-language full-text articles were included. Eligible studies comprised randomized controlled trials, observational studies, systematic reviews, meta-analyses, relevant experimental research (in vitro and animal models) and official guidelines. Given the heterogeneity of the available literature, a narrative synthesis was performed, with findings organized into thematic domains including Urinary Tract Diseases classification and definition, *Escherichia coli* virulence, Om-89 effectiveness and worldwide recommendations, and new immunological rUTI prophylaxis discoveries. No meta-analysis was conducted.

E.coli virulence

Uropathogenic Escherichia coli is the predominant species of pathogens in UTIs. The pathogenesis of UPEC relies heavily on its ability to ascend from the gastrointestinal tract and colonize the specialized host epithelium of the bladder. A critical mechanism driving this process is bacterial adherence to the uroepithelium, which prevents the mechanical clearance of bacteria through urination. This attachment is mediated by type 1 pili, specialized surface structures that display an adhesin protein known as FimH at their fimbrial tips. The lectin binding domain of FimH (FimHLD) specifically attaches to mannosylated glycoproteins present on mammalian bladder surfaces [9,10]. Once anchored, UPEC can invade the bladder epithelial cells and proliferate intracellularly, forming dense intracellular bacterial communities (IBCs). These IBCs act as dormant, protected reservoirs that shield the bacteria from host immune detection and severely limit the intracellular penetration of conventional antibiotic treatments. When antibiotics are discontinued or host conditions shift, these

reservoirs can reactivate, leading to persistent bacterial shedding and high rates of clinical recurrence.

Uro-Vaxom (OM-89) as an Immunoactive Prophylactic Strategy

The first clinically safe immunotherapeutic Uro-Vaxom (OM-89) was registered in Germany and Switzerland in 1987. This bacterial extract contains six milligrams of lyophilized bacterial lysates from eighteen strains of *E. coli* which stimulates immunological response [3]. It promotes T-lymphocytes proliferation and dendritic cell formation, induces interferon release and raises IgA level in the urine. It's administered orally for 90 days and can be prescribed from the age of 4. Data about OM-89 usage during pregnancy are limited and insufficient, therefore it's not recommended in this group of patients [11]. Both safety and efficacy of this immunomodulator have been widely investigated over the years. The early review by Kurt G. Naber and co from 2009 [12] summarised the first 7 trials revealing that using OM-89 decreases the number of rUTI by almost 36%, additionally reducing dysuria and possibly leukocyturia or bacteriuria after 6 months of usage. OM-89 efficacy was also confirmed in a study involving menopausal women with 65% reduction in rUTI. However, a significant reduction in this treatment option was noticed in regard to diabetics [13]. That data resulted in OM-89 prophylaxis being recommended by the European Association of Urology (EAU) [4], Colegio Mexicano de Especialistas en Ginecología y Obstetricia (COMEGO) [14], Sociedad Española de Enfermedades Infecciosas y Microbiología Clínica (SEMIC) [15], Swiss Society of Gynaecology and Obstetrics (SSGO) [16] and Arbeitsgemeinschaft der Wissenschaftlichen Medizinischen Fachgesellschaften (AWMF) [17] guidelines by 2022. OM-89 was also recommended in 2024 by Guidelines of the Polish Urological Society, Polish Society of Gynaecologists and Obstetricians and the Polish Society of Family Medicine [1] on the diagnosis, therapy and management of postpartum lower urinary tract infections. However, that trend has since declined. The literature review conducted in 2025 [5] summarized existing trials showcasing that OM-89 efficacy exceeded the prior estimates and reduced rUTI and dysuric symptoms from approximately 34% to over 80% compared to placebo. Thus the review also stated that the presented studies were heterogenic in outcome reporting, varied in follow-up durations, and lacked consistency in how recurrence was defined and quantified across studies. The lack of a standardized effect size across trials made estimating Uro-Vaxom's true efficacy impossible. The most recent study of Uro-Vaxom was conducted by Sao Paulo's physicians [18] later that same year and highlighted a significant decrease in OM-89 efficacy, questioning its reliability as a preventive rUTI treatment. International guidelines as it comes

to prescribing immunological rUTIs prevention have changed significantly in recent years. The European Association of Urology that used to strongly encourage immunoactive prophylaxis for rUTIs with quality of evidence 1a in 2022 [19], has downgraded the previous strong recommendations to weak in the newest May 2026 edition. The decision was based on low certainty of the available data. The American Urological Association (AUA) [20] has not recommended any urological vaccine due to insufficient quantity and quality of evidence from the beginning. The 2025 Guideline update for Recurrent Uncomplicated Urinary Tract Infections maintains the thesis about the lack of reliable conclusions regarding immunomodulants effectiveness due to the small number of trials, imprecise estimates, and methodological shortcomings.

Future high-quality randomized controlled trials with larger, gender-balanced cohorts and standardized outcome measures are essential to further investigate OM-89 role in recurrent Urinary Tract Infections prophylaxis.

Stabilized FimH Subunit Vaccines

While whole-cell lysates like Uro-Vaxom offer broad immunostimulation, modern vaccine design has increasingly pivoted toward highly targeted, parenteral subunit antigens, specifically utilizing the structural components of the FimH adhesin protein. The primary mechanism behind a FimH-based vaccine is the generation of antibodies that directly block the lectin binding domain, effectively neutralizing the bacteria's ability to adhere to uroepithelium [21]. There have been many studies conducted on non-human models to find the proper structure and dosage for immunization with recombinant FimH_{LD} or full length FimH in complex with FimC (FimCH) [22-31] with some promising results for further evaluation human trials such as FimH-DSG™. Moreover, there was already Phase 1 study human trial carried out with results submitted in 2025 by Eldridge and co [9]. It studied safety and immunogenicity of four doses of FimCH combined with a TLR4 adjuvant. This not-randomised study was conducted at six clinical sites enrolling in women ages 21 to 64 from December 2013 to November 2015. The vaccine used in this study consisted of FimCH (antigen) and Phosphorylated HexaAcyl Disaccharide (PHAD®; adjuvant). Intramuscular injections were administered on days 1, 30, 90, and 180 with the majority of adverse reactions being local tenderness, mild pain and burning that resolved within seven days. No severe systemic reactions were observed. Due to the favorable safety profile and immunogenicity results, the adjuvanted FimCH vaccine is to be further examined in a double-blind, randomized, placebo-controlled phase 2 study.

Discussion

Recurrent urinary tract infections (rUTIs) represent a significant clinical burden, particularly among adult and postmenopausal women, drastically diminishing quality of life and generating substantial socio-economic costs. For decades, the therapeutic mainstay for both acute episodes and prevention has relied on empirical antimicrobial therapy. However, the escalating global crisis of antimicrobial resistance, exacerbated by uropathogenic *Escherichia coli* strains acquiring multidrug resistance, has enforced changes in therapeutic approach. Current research and clinical guidance increasingly emphasize non-antibiotic strategies, with particular focus on immunotherapeutic interventions, targeted bacterial adhesin vaccines, and competitive inhibition of bacterial adherence. The high rate of UTI recurrences following appropriate short-course antimicrobial treatment is rooted in the complex pathogenesis of UPEC, especially the FimH adhesin tightly anchoring bacteria to the bladder uroepithelium. This adherence enables intracellular invasion, where intracellular bacterial communities develop, protected from both host immune system and conventional antibiotics due to the lack of adequate intracellular distribution. When antibiotics are withdrawn or host physiological conditions alter, bacteria can re-emerge, triggering clinical relapse. To break this recurrent cycle without fueling further resistance, therapeutic modalities must target bacterial virulence factors or enhance host mucosal immunity rather than rely solely on bactericidal compounds.

Re-evaluating Whole-Cell Lysates: The Evolution of OM-89 (Uro-Vaxom)

Oral immunoprophylaxis with OM-89 (Uro-Vaxom)—a lyophilized bacterial extract derived from 18 UPEC strains—has historically served as a core non-antibiotic prophylactic option. OM-89 acts as a broad mucosal immunomodulator, activating dendritic cells, stimulating T-lymphocyte proliferation, inducing interferon production, and promoting urinary IgA secretion. Early clinical trials and reviews, such as those summarized by Magistro & Stief (2019) and Naber et al. (2009), established its safety and demonstrated significant clinical efficacy, reducing rUTI recurrence rates by roughly 36% to over 60%. These findings led to strong recommendations across multiple guidelines, including the European Association of Urology (EAU), German AWMF, and Polish urological societies.

However, recent critical appraisals and systematic literature reviews have highlighted critical methodological limitations in earlier evidence. While updated meta-analyses confirm that OM-89 significantly reduces UTI recurrences and antibiotic consumption compared to placebo, they also emphasize substantial study heterogeneity, variable follow-up durations, and inconsistent definitions of infection recurrence across historical trials. Furthermore, real-world comparative

studies and trial sequential analyses have raised questions regarding the magnitude of OM-89's efficacy, particularly in specific sub-populations like diabetic patients. Consequently, major international bodies have revised their position: the EAU downgraded its endorsement of immunoactive prophylaxis from a strong to a weak recommendation in its 2026 guidelines, citing low certainty of available evidence, while the American Urological Association (AUA) continues to refrain from endorsing bacterial vaccines until higher-quality data become available.

Target-Specific Subunit Vaccines: FimH Innovations

To overcome the variability and non-specific nature of whole-cell lysate preparations, contemporary research has pivoted toward parenteral, structurally targeted subunit vaccines. FimH-based vaccines aim to elicit potent systemic and mucosal neutralizing antibodies against the lectin-binding domain of the FimH adhesin, directly disrupting bacterial docking to the uroepithelium.

Preclinical evaluations and recent Phase 1 human clinical trials involving bi-subunit formulations (such as FimCH combined with a synthetic TLR4 adjuvant like PHAD®) have demonstrated a favorable safety profile and robust antigen-specific immunogenicity. Systemic administration induces neutralizing antibodies capable of blocking UPEC attachment across diverse strains without causing severe systemic adverse events. By neutralizing the initial binding event, these targeted vaccines prevent intracellular invasion and IBC formation altogether, addressing the root cause of persistent reservoirs.

Competitive Anti-Adhesion Strategies: Oral FimH Inhibitors

In parallel with active immunization, competitive adherence inhibition offers an alternative non-antibiotic pathway. D-mannose has long been used as a natural prophylactic agent that binds to FimH, occupying its lectin domain and preventing urothelial attachment. Recent advancements have focused on optimizing this strategy through novel synthetic FimH antagonists, such as oral 1-deoxymannose derivatives. Pharmacokinetic and excretion studies demonstrate that orally administered 1-deoxymannose yields high urinary concentrations, allowing direct, competitive binding to UPEC in the bladder lumen. This dual pathway—combining competitive oral FimH inhibitors for immediate anti-adherence with targeted FimH vaccines for long-term immunologic protection—represents a highly promising frontier in rUTI management.

Conclusion and Future Perspectives

The clinical management of recurrent UTIs in women is transitioning away from long-term low-dose suppressive antibiotics toward multimodal, non-antibiotic prophylaxis. While OM-89 remains a widely utilized immunomodulator, shifting international clinical guidelines underscore the critical need for well-designed, adequately powered, randomized controlled trials with standardized outcome metrics to definitively establish its modern therapeutic role. Concurrently, highly specific target-driven approaches—notably adjuvanted FimH subunit vaccines and synthetic FimH_{LD} competitive inhibitors—present a compelling, mechanistically sound strategy to eradicate UPEC reservoirs, lower recurrence rates, and support global antimicrobial stewardship goals.

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Artificial Intelligence use declaration

AI tools were used only to refine the language and improve text readability. The authors retain full accountability for the academic integrity of the work. The conceptual design, literature review, data analysis, and final drafting were executed solely by the authors.

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